CVIN-SC: The Work So Far — the research side

Oct 31, 2026 (freeze) · Nikhil Prakash · v0-3 · **Companion to the code-side document of the same title. That document covers the applied system; this one covers the research corpus, the specification and the evaluation of record that the thesis is built from.**

What this is

CVIN-SC asks what an identity layer for connected and autonomous vehicles must minimally provide, whether the blockchain identity standards now in print provide it, and whether the answer survives the move from vehicles to software agents. The research side answers those questions by derivation, by a standards inventory and by a pre-registered evaluation of eight implementations of one interface; the code side (the companion document) answers a different and complementary set by building and measuring a full vehicular system. Every research-side number in the thesis is tied to a committed trace in the hub repository, `nikhilprakash24/cvin-sc` (private; access on request), and every claim carries one of three status words: *verdict* (gate-passed, resolves to a committed trace), *expectation* (predicted, or from an unrun or superseded instrument) or *analysis* (derived without measurement).

This document lists what the research side does that I have not found done elsewhere, what it found, where it was frozen, where each piece lives, and where the thesis stands.

What this side does that I have not found elsewhere

Each item is stated for what it is: derived, measured or argued, with its status, and positioned against the literature in Chapter 2. Priority claims invite counter-examples, so none is phrased as "first". Throughout these documents, "panel", "lens", "examiner" and "review" name roles performed by the AI assistant under my direction; no one other than me has yet reviewed any of this work.

1. **A derived, not assembled, minimal identity interface.** `IMinimalSSI`, twelve functions, derived from a 29-operation universe assembled from the W3C DID Core and VC Data Model normative operations and Allen's ten principles, under a published clustering rule (CR-1…CR-3), by a two-part necessity argument. The result is stated as a *sufficient and irredundant cover*, explicitly not the unique minimum; alternatives under a coarser rule fall to nine or ten functions without touching the credential operations. The derivation is mechanically checkable by re-running two elimination tests on the committed operation map. *(analysis)*
2. **The necessity partition and "no thirteenth function".** Three functions are principle-necessary (`createDID`, `resolveDID`, `createPresentation`), five operation-necessary, four conventionally necessary — and all the conventionality sits in key management. Seven principles are discharged on the function layer and three off it, each at a named site. Presentation *verification* folds into `verifyCredential`; there is no thirteenth function. *(analysis; CEV-16…18)*
3. **A closure relation for a complete identity system.** "Complete" is defined as *closed under the stated lifecycle transitions over the exhibited object set*, with a falsification

condition, a role table (issuer, holder, verifier, VDR, controller, delegate) and a lifecycle closure table. The analysis found one closure failure — delegate revocation before expiry — and resolved it by Specification Amendment F11, corroborated in registry code. It is the only place the analysis changed the specification. *(analysis, falsifiable)*

4. **`createPresentation` as the sole discharge of Consent and Minimization — three claims that fail independently.** Necessity (analytic, §2.3); absence from the ERC register (a dated census, not a literature search: provisional count over 30 ERCs, none native, 17 partial, 13 absent); and per-option realisation (on every EVM option the function is a replay-protected anchor, scored *absent* by the principle-preservation rule). *(analysis · expectation until the census is dated · verdict for ERC-1056)*
5. **A standards inventory on one footing.** Seventeen standalone options in six families (ERC registries, tokens, accounts, agent registries, the Indy/Aries and Identus sub-layers of Family E, and the Family F composite O18), each profiled against the twelve functions at a freeze date, with a standards-symmetry table (DID operations ↔ Set A; VC lifecycle ↔ Set B), a Family E sub-layer decomposition and a Family G census of ERC-1056 industry lineages. *(framework; inventory partly issued)*
6. **A pre-registered evaluation protocol, with its disclosure.** Fixed by commit `0a10831` on 15 October 2026 before any option other than ERC-1056 had been ported: measurands, N = 5 per gas arm and N = 30 per timing, seven void rules, a Full/Partial/Absent rubric with four tests, self-audit controls. The eight-option runs made after it but not under it are admitted by ruling D-R7 with the deviation stated in the thesis's own voice (a dated note in §2.5); the eleven ERC-1056 runs that pre-date the protocol, two within ten minutes of the commit, are listed by a dated disclosure paragraph drafted on 5 November (v1-2; trace keys to be confirmed at source before it is appended); gas was then re-measured under the protocol (D-R8: 128 of 128 cells at N = 5, every repetition identical; deployment gas at N = 5 on all fourteen rows). *(method; verdict for the re-measurement)*
7. **One reference result that reproduces to the digit.** ERC-1056 passes 12/12 under a pinned compiler (solc 0.8.24), and eleven of its twelve transaction gas values are identical across three runs over four and a half months and two compiler states; the twelfth varies for one identified reason (an unseeded challenge, −12 gas per zero byte). The 17,100-gas difference between the first and later identity writes is exactly one fresh-slot SSTORE (22,100 − 5,000), visible inside one trace on every option. *(verdict)*
8. **Eight native ports of one interface, admitted by one unmodified suite.** ERC-1056, ERC-721, ERC-725, ERC-725+735, ERC-1155, ERC-4337, ERC-8004 and Indy-on-Besu each pass the same twelve tests; scenario code written above the interface runs unchanged on all eight. Coverage and gas order the standards differently (native functions of twelve: 9, 9, 8, 6, 6, 5, 3, 2; creation gas from ≈0.16 M on light registries to ≈1.63 M on account-per-identity designs). Eighteen interface-level findings change the specification rather than the numbers: entitlement, content binding, supersession, controller transfer and early delegate revocation lie outside the twelve functions. *(pre-protocol, admitted; findings as existence and impossibility results)*
9. **Two independent analyses that draw the same boundary.** An on-chain/off-chain placement procedure (eight statement classes, seven placement forms, four tests in order — act, personal, equivocation, Tier-2 read) puts Set A state on-chain and Set B content off-chain and anchored; the minimality derivation drew the same Set A / Set B

line for different reasons. Hash-only anchoring is both the default and the cheapest form; throughput is shown by arithmetic not to be the binding constraint, which sends the argument to latency and privacy. *(analysis)*

10. **The agentic generalisation as an interface claim.** A mapping φ from vehicular to agent principals that preserves the interface and not the trust model: five functions transfer exactly, seven modified, none absent; nine non-correspondences and two design rules named; an ERC-4337/ERC-7579 smart-account wrapper specified module by module with a pre-declared validation ladder. Secure identity-to-identity messaging for agents was the subject of the three memos that re-opened this programme (project listing dates them 25 January 2026; any precedence claim is to be made from dated artefacts, not from this sentence). *(analysis; wrapper at specification only)*

Beneath these, a corrections ledger: the research side found and corrected figures that earlier material you have seen carried wrongly — most importantly the BSM broadcast rate (10 Hz nominal, not 1,000–2,000 per second; the larger figure is receive-side load) and the Groth16 on-chain verification cost (≈181,000 + 6,150 per public input after EIP-1108, not ≈350,000). These are reported, not silently replaced.

Headline results

Question	Result	Status · where
RQ1 derivation	Twelve functions as a sufficient and irredundant cover of R = W ∪ P under CR-1…CR-3; Set A (8) / Set B (4); partition 3 / 5 / 4	analysis · §2.3 v2.0, App. M-1
RQ1 closure	Closed under stated transitions over the exhibited object set; one gap found and amended (F11)	analysis, falsifiable · App. M-2
RQ2 reference option	ERC-1056: 12/12; `createDID` adapter write 178,935 first / 161,835 after; `issueCredential` 130,163; `createPresentation` anchor 62,458; eleven of twelve values reproduce to the digit across three runs	verdict · trace `i2i_01_ERC1056_1792279261701.jsonl` @ `cvin-sc@2ae7989`
RQ2 eight options	All 12/12; Set A fully native on ERC-1056 only; Set B native on no EVM standard except claims or tokens without expiry; `createPresentation` an anchor everywhere, off-ledger on Indy	pre-protocol, admitted (D-R7) · `coverage_matrix.csv`; Phase 1 record

Question	Result	Status · where
RQ2 protocol gas	8 options × 9 write functions × first-write / rewrite, N = 5, three warm-ups, seeded order, manifests; 128/128 cells with identical repetitions; deployment gas N = 5 on 14 rows (ERC-1056 1,216,044 … ERC-8004 5,363,688)	verdict · `verification/2026-10-22_phase_b_protocol_gas/`, `2026-10-23_deploy_gas_n5/`
RQ2 census	30 ERCs read at source: N = 0, P = 17, X = 13	expectation (single rater; date and second rater owed) · census v0-1
Placement	Boundary = Set A / Set B; hash anchor cheapest; throughput not binding	analysis · P2, HAA-1, cost thread
Latency framing	Tier boundaries adopted (100 / 500 ms), not measured; Tier-2 path carries only reads in every scenario; "Tier-2 compliance is not a ledger property"	analysis + illustration · §2.4, §4.5–4.6, `tier_coverage.md`
Security	Revocation window as a functional-safety budget; validator–issuer overlap Ω as the strongest novel security result; 35-entry threat catalogue with 33 variables	analysis · T-1, G-C1
RQ3 agents	φ preserves interface, not trust model; 5 exact / 7 modified / 0 absent; wrapper specified, no rung reached	analysis · §5.2, App. I

Nothing on this side is stated at a status it has not earned. Where a result is an illustration (the one-vehicle corridor scenario: 26 invocations, nine principals) it is labelled as one.

The freeze point

The research side froze with the code side on 31 October 2026, for the same reason: the field is moving (VC Data Model 2.0 at Recommendation since May 2025; DID 1.1 at Candidate Recommendation since March 2026; ERC-4337 and ERC-8004 still settling; EDPB guidance of July 2026 breaking the "GDPR by hash-anchoring" sentence that earlier drafts carried) and the work has moved with it. The frozen state is the hub repository at `d9f9fa3` (31 October, the unified-thesis ruling D-R9); the project store stood at 168 filed documents when this was written (4–5 November), with only reporting instruments added since the freeze. What arrives

after the freeze — a new standard, a new ruling, a new draft — goes to the programme around the thesis unless it corrects a number of record.

The three sides, and which this document is

The past twelve months have three sides. The **implementation side** is the code-side companion document. The **full pass and exploration of March to May 2026** — more than a hundred pages of notebooks re-examining the CVIN work from first principles — is not in either repository or the project store and is listed here so the shape is visible; I will present it myself. This document is the **written research side**: the derivation, inventory, protocol, analysis threads, drafts and governance that the project store and the hub repository hold.

The corpus, by category

The research side lives in two places. The **project knowledge store** (claude.ai project "CVIN Thesis") holds 168 filed documents, versioned `v<major>-<minor>`, nearly all with a first and a second pass. The **hub repository** `nikhilprakash24/cvin-sc` holds the sandbox of record, seven verification folders, the chapter drafts and a document corpus imported from the store in thirteen categories (`docs/00_governance` … `docs/12_spoke_import`). Start with the hub's `STATUS.md` and `docs/INDEX.md`.

Category	What it holds	Size · where
Specification and derivation	The formal function specification (Jan 2026); the hardened §2.3 derivation v1.0 → v2.0; minimality hardening G-A1; coverage checkability G-A2; the operation universe and elimination trace; the closure contribution and Appendix M-2; function cards F1–F12 with pseudocode	14 filed docs; hub `docs/01_specification/`, `docs/10_drafts/`
Evaluation of record	The sandbox v1.1 (June baseline, recovered 15 Oct) and v1.3 (eight options, 63 traces, 133 passing); seven verification folders; the pre-registered §2.5 and its deviation note; protocol gas and deployment gas; the Phase 1 consolidated record (every number pinned to a path and line)	hub `sandbox/`, `verification/`; Phase 1 record G-P1 v0-2
Standards and families	Building Paths v1.1 → v4.0; Family E expansion v2.0; the Family G lineage census; the	13 filed docs; hub `docs/09_content_sources/`,

Category	What it holds	Size · where
	provisional ERC presentation census; the MOBI VID requirement inventory, evidence register and implementation-options plan; the vehicle-identity populations classification	`docs/07_audits_plans/`
Analysis threads	Placement (plan of attack, analytical core, hybrid anchored architecture, pass record); security and privacy campaign G-C1 and the T-1 threat catalogue; empirical rigour protocol G-B; agentic validation G-E1; cost thread G-CO1 (14 cost classes, payload ladder); roots thread G-R1 (technology lineages to the stack, 51 + 42 = 93 breaking points across Phases 1–2, panel review R9); the hybrid-data campaign HDA (18 documents, ≈166,700 words, one measured experiment)	12 filed docs + HDA in hub `docs/11_hda_campaign/`
Governance and single sources of truth	Scaffolding and completion plan v0.1 → v0.4; the consolidation audit and production manual v2.0; the master writing plan; the thesis content registry v12.0 → v13.1; chapter sectioning documents for all five hub chapters; accuracy ledger and terminology freeze; gap taxonomy (the G-code scheme); the crux defence compendium X1–X6 (+X7)	28 filed docs; hub `docs/00_governance/`, `02_thesis_masters/`, `03_sectioning/`, `04_gap_modules/`
Verification passes	Passes 1–5 over the corpus (verification dossier, deep verification and the claim–evidence–verdict ledger, CAV/IoV grounding, Session A reconciliation)	8 filed docs; hub `docs/05_verification_passes/`

Category	What it holds	Size · where
Thesis prose	Hub drafts of record: Ch. 1 v1-1, Ch. 2 complete (§2.0/2.1/2.6, §2.2+2.4, §2.3 v2-0, §2.5 v1-1), Ch. 3 v0-9 (partial), Ch. 4 complete (§4.0–4.9), Ch. 5 v1-0 — ≈60,200 prose words; a reading copy of 23 October; the G-RIM results and methodology drafts; Take 1 (4 Nov, ≈27,000 words)	hub `docs/10_drafts/`; project + chat
Programme views	Programme Atlas (every chat since 10 Feb 2026 by lane, with outputs and what to solidify); Mission Control dashboard; unfiled-outputs register (61 output sets not yet filed, 21 at priority 0); the precursor report and supervisor cover briefs of 22 Oct	6 filed docs
Vision and future programme	Full-send research dossier and bold-claims ledger (no forward claim above specified composition); IoV possibility compendium and the identity-transaction base catalogue; the necessity → machines → agents argument (G-WHY); societal integration; the primitives-first argument in three versions	20 filed docs; enters the thesis as future work and one worked example
Drafting programmes	Grounding draft G-GD1 (10 phases, planned); supervisor draft G-SD1 (10 stages, origin tags, provenance ledger, AI-use brief — planned); the takes programme G-T1 (Take 1 delivered; Takes 2–4 roadmapped)	8 filed docs

The documents a reader will want most:

Document	Why read it
Precursor Report, July–October overhaul (G-PRE v0-2, 22 Oct)	The research side in one report: five changes, the five-move argument, the evidence register, the corrections ledger, the cruxes, the four results to examine first
Phase 1 Consolidated Record (G-P1 v0-2, 24 Oct)	Every research-side number with a repository path and line; the admissibility classes A–D; the findings catalogue; the drift and defect register
Thesis Ch. 2 §2.3 Derivation v2-0 (hub)	The strongest material in the corpus
Thesis Ch. 2 §2.5 Methodology v1-1 (hub, pre-registered 15 Oct)	The protocol the evaluation is held to, and the disclosure of where it was not
Hub vs Spoke Comparison and Merge Proposal v0-1 (hub, 30 Oct) and ruling D-R9	How the two sides become one thesis: overlaps, seventeen conflicts, the unified outline, the numbers-of-record rule
Take 1 Panel Note, Pass Log and Roadmap (G-T1 v0-2, 4 Nov)	What a continuous draft of the hub thesis looks like, what the panel changed, and the ranked list of what to solidify

The thesis drafts, and what comes next

The hub holds five drafted chapters of the specification-first thesis (≈60,200 prose words, 23 October reading copy), and a lift-and-assemble draft in my own voice that reads continuously from abstract to conclusions (Take 1, ≈27,000 body words, 93 placeholders counted, 4 November). The code side holds seven drafted chapters of the applied system (≈45,000 words by its own count). Ruling D-R9 of 30 October makes these one thesis: seven chapters, about 68,000 words, hub RQ1–RQ3 principal with the code side's questions as sub-questions, the applied system as Chapter 6, hypotheses H1–H6 kept as the code side's reported expectations, and the two sides' gas never mixed in one table.

Chapter (unified)	State
1 Introduction	Hub v1-1 drafted; contributions table audited; RQ adoption marked ⟦RQ-ADOPTION⟧ pending your view
2 Background and related work	Hub §2.0–2.2, 2.4, 2.6 drafted; code-side literature review to be merged with absence claims resolved
3 IMinimalSSI specification	Hub §2.3 v2-0, to be promoted to its own chapter unchanged
4 Methodology	Hub §2.5 v1-1 (pre-registered) + code-side Ch. 3; roles of N = 5 (protocol) and N = 30 (determinism check) kept distinct
5 Standards inventory and evaluation	Hub Ch. 3 v0-9 (§3.2 census pending) + Ch. 4 §4.2–4.6

Chapter (unified)	State
6 Vehicular instantiation	Code-side Ch. 4–5 condensed; native-analogue gas in its own labelled table
7 Discussion and conclusions	Hub Ch. 5 v1-0 + code-side Ch. 6–7 trimmed
Appendices	Operation universe, closure table, agent wrapper spec, risk register, protocol gas tables, source register

Next, in order: your reply on the five decisions; the AI-use disclosure and agreement with the authorship timeline (the provenance-tagged supervisor draft cannot start without it); the eleven pre-registration traces listed by date in a §2.5 amendment; the ERC census executed and dated with a second rater; the migration steps of D-R9 (crosswalk, one citation system, locked-wording sweep on the imported text, then chapter assembly); the file-in of the 61 unfiled output sets so that every reconstruction in Take 1 becomes a lift.